

01:XXX:XXX - Homework n

Pranav Tikkawar

September 22, 2026

Contents

1 Problems	2
------------	---

1 Problems

Example (3 Coloring of NxN). what is the number of all 3 colorings of an NxN grid such that no two adjacent cells have the same color? LB = $\sqrt{2}^{n^2}$
UB = $3 * 2^{n^2}$

$$N(n) \sim \frac{4^{\frac{3}{2}n^2}}{3}$$

this number is the squareice constant
Ice Problem Elliot Lieb

Example (Domino Tilings of NxN). what is the number of ways to tile an NxN grid with 2x1 dominoes? Not possible if N is odd
Im thinking LB is $(\frac{N}{2})^2$
UB is $N!$

$$N(n) \sim$$

For $N(8) = 12988816$
We know for N even it exists

Example (Magic Square). MS for n=2 does not exist

Proof. Let the square be

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

then we have $a + b = c + d$ and $a + c = b + d$ and $a + d = b + c$

then $a = d$ and $b = c$ and $a = b = c = d$

so all numbers are equal which is not possible since we want distinct positive integers. □